

Antilithic effects of extracts from different polarity fractions of *Desmodium styracifolium* on experimentally induced urolithiasis in rats

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Abstract *Desmodium styracifolium* (*D. styracifolium*) has been widely used in traditional Chinese medicine for the treatment of urolithiasis. This work was undertaken to investigate extracts from different polarity fractions of *D. styracifolium* for possible antilithic effects as well as antioxidant potential to explore the underlying phytochemically active constituents of this plant. The extracts of *D. styracifolium* were divided into four different polarity fractions by petroleum ether (Fr. PE), chloroform (Fr. CH), ethyl acetate (Fr. EA), and *n*-butyl alcohol (Fr. NB). The antilithic and antioxidant effects were evaluated and compared in vivo on an animal model of calcium oxalate (CaOx) urolithiasis, which was established by administration of 1 % ethylene glycol along with 2 % ammonium chloride in drinking water for 28 days. A total of 60 male Sprague–Dawley rats were randomly divided into six groups: normal control group, lithogenic group, and four different polarity fractions of *D. styracifolium*-treated groups. At the end of the study, urine, blood, and kidney tissue samples were all collected for evaluation. Among the four polarity fractions of *D. styracifolium* extracts, the Fr. PE and Fr. NB treatment significantly reduced the

CaOx crystal deposition in kidneys, prevented the renal toxic changes like pH, Cr, and BUN. In addition, Fr. PE and Fr. NB treatment significantly decreased urinary excretion of oxalate along with an increase of citrate excretion. The increased amounts of malondialdehyde and decreased activities of superoxide dismutase, catalase, and glutathione peroxidase were detected in lithogenic group. *D. styracifolium* extracts treatment prevented the oxidative stress changes especially for the Fr. PE and Fr. NB extracts. In conclusion, our data suggest that the extracts from *D. styracifolium* possess the antiurolithic activity, possibly mediated through the inhibition of CaOx crystal aggregation as well as the alleviation of oxidative injury in the kidney, and the Fr. PE and Fr. NB extracts are the active fractions of *D. styracifolium* extract.

Keywords *Desmodium styracifolium* · Urolithiasis · Calcium oxalate · Antioxidant · Rats

Introduction

Urolithiasis, the formation of urinary stone, is a prevalent disorder which mainly affects the places with high humidity and temperature geographically, and is associated with high rate of recurrence [1, 2]. Hyperoxaluria is one of the major risk factors for renal stone formation, and approximately 70–80 % of renal stones are composed of calcium oxalate (CaOx) crystals [1]. Despite the evolved understanding mechanisms of crystallization and stone formation, current effective medical treatment advance remains to be modest because of the slow progress in determination of stone formation pathophysiology [3]. Various medical and surgical techniques have been developed to effectively manage kidney stone

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disease. However, application of these techniques is restricted due to the cost and complications [3]. Moreover, the high recurrence rate of kidney stones remains an important unsolved problem [1]. Therefore, the development of new agents against urolithiasis is still urgently needed. Due to significant synergic therapeutic effect and relatively low toxicity of the herbal medicine, increasingly considerable attention has been given to the antilithic natural products in the recent years [4–6].

Desmodium styracifolium (Osbeck) Merr. is a plant member from the family of genus *Desmodium* (Fabaceae). The aerial parts of *D. styracifolium*, called “Guang Jinqiancao” in China, are a well-known traditional Chinese medicine, and have been widely used for treatment of renal stones due to its heat-clearing and diuretic properties [7]. In vitro crystallization study has confirmed the prophylactic effect of *D. styracifolium* in the formation of calcium oxalate crystals [8, 9]. Further in vivo study subsequently demonstrated that the aqueous extracts from *D. styracifolium* inhibited CaOx stone formation by rising urinary citrate excretion, decreasing urinary calcium, diuresis, and enhancing antioxidative effects in rats [10]. We have also previously reported the antiurolithic and antioxidant activities of *D. styracifolium* aqueous crude extract in vivo, which is consistent with the above [11, 12]. However, the potential active phytochemical constituents of this plant have not yet been fully understood.

Thus, the present study was performed to investigate the antiurolithic and antioxidant effects of extracts from different polarity fractions of *D. styracifolium* on ethylene glycol-induced calcium oxalate renal stones in rats.

Materials and methods

Plant material

The aerial part of *D. styracifolium* were procured from Kangmei Pharmaceutical Co. Ltd. (Lot number: 091107141), and was authenticated at the School of Chinese Materia Medica, Guangzhou University of Chinese Medicine, Guangdong, P.R. China. A voucher specimen (ZYXY-D. STYRACIFOLIUM-0010) was deposited at the herbarium for further reference.

Extracts preparation

The air-dried aerial part of *D. styracifolium* (5.75 kg) was powdered, and extracted with 70 % aqueous-ethanol (40 L × 3) three times (60 min each) under reflux. The obtained extract was filtrated and concentrated under reduced pressure on a rotary evaporator to leave a residue. The residue was suspended in water (3 L), and then

subjected to successive extraction using petroleum ether (3 L × 3), chloroform (3 L × 3), ethyl acetate (3 L × 3), and *n*-butyl alcohol (3 L × 3) in the order of increasing polarity. The four extracts were concentrated by distillation under reduced pressure, yielding 9.0 % (w/w), 4.2 % (w/w), 2.4 % (w/w), and 6.6 % (w/w) for petroleum ether (Fr. PE), chloroform (Fr. CH), ethyl acetate (Fr. EA), and *n*-butyl alcohol (Fr. NB), respectively.

Animal models

A total of 60 male Sprague–Dawley rats, weighing approximately 180–220 g, were purchased from Guangdong Medical Laboratory Animal Center (Foshan, Guangdong). All rats were kept in a specific pathogen-free animal facility under controlled conditions at the temperature (24 ± 2 °C), humidity (55 ± 15 %), with a 12-h light–dark cycle. The experimental procedures were approved by Animal Care and Use Committee of Guangdong Provincial Hospital of Chinese Medicine and were in accordance with recommendation of the National Institutes of Health Guide for the Care and Use of Laboratory Animals. Animal model of CaOx kidney stone was established orally by administration of 1 % (v/v) ethylene glycol (EG) and 2 % (w/v) ammonium chloride (AC) 2 mL in drinking water for 4 weeks [13]. To investigate the antilithic effects of extracts from different polarity fractions of *D. styracifolium* on experimentally induced urolithiasis, rats were randomly divided into six groups: (1) control group; (2) lithogenic group: 1 % EG + 2 % AC; (3–6) *D. styracifolium*-treated group: 1 % EG + 2 % AC + 2.4 g/mL (pharmacognostic concentration) Fr. CH/Fr. EA/Fr. PE/Fr. NB, respectively [14]. The doses of fractions from *D. styracifolium* were calculated according to their yields in the extraction and preparation, which were all equivalent to 2.4 g/mL of unprocessed *D. styracifolium*. They were 2.16 g/kg for Fr. PE, 1.008 g/kg for Fr. CH, 0.576 g/kg for Fr. EA, and 1.584 g/kg for Fr. NB, respectively. All animals had free access to drinking water and regular chow. The consumption of food and water intake was recorded every day, and the weight of rats was checked weekly to check the growth.

Samples collection and analysis

At the end of 28 days of treatment, the rats were placed into a metabolic cage individually to collect 24 h urine samples. After determination of urine volume and pH, the urine samples were acidified to pH 2 with 5 M HCl and centrifuged at 1,500 rpm for 10 min to remove debris and the supernatants were stored at –20 °C until analysis. The rats were anesthetized intraperitoneally with 10 % chloral hydrate, and then the blood was collected through inferior vena cava. After centrifugation at 4 °C, the serum separation was

transferred into clean tubes and stored at -20°C for oxidative stress analysis. Both the kidneys were excised and rinsed in ice cold physiological saline. The right kidneys were fixed in 10 % neutrally buffered formalin for hematoxylin and eosin (H&E) staining and microscopic examination of CaOx crystals. The left kidneys were stored at -20°C for oxidative stress analysis.

In acidified urine samples, calcium (Ca^{2+}) and magnesium (Mg^{2+}) contents were determined by using commercially available kits. For detection of urinary oxalate and citrate, analyses were performed by ion chromatography method using a Dionex ICS-2500 gradient system equipped with a 0.4×25 cm IonPac AS11-HC anion analytical column and a 0.4×5 cm IonPac AG11-HC guard column. While in serum, creatinine and BUN were estimated with kit-based methods using automatic biochemistry analyzer. Malondialdehyde (MDA), superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GSH-Px) were examined by the commercial kits from the Nanjing Jiancheng Biochemistry according to the manufacturer's protocols. Crystal deposition within the kidneys was determined using a semiquantitative scoring method. Briefly, the score of crystal deposition was expressed by the product of distribution score multiplied by extent score, as shown in Table 1.

Table 1 Crystal deposition score system

	Cortex 1 pt	Medulla 2 pts	Papilla 3 pts	Pelvis 4 pts
None 0 pt	0	0	0	0
Little 1 pt	1	2	3	4
Some 2 pts	2	4	6	8
Many 3 pts	3	6	9	12

Statistical analysis

The data were expressed as mean $\pm$ standard deviation (SD) and were analyzed with one-way ANOVA followed by Newman–Keuls post hoc test for multiple comparisons (GraphPad Prism 5.0, San Diego, CA, USA). The crystal deposit scores were analyzed by the non-parametric Kruskal–Wallis test for inter-group comparisons and by the Wilcoxon rank-sum test (Mann–Whitney *U*) for pairwise comparisons. The *P* values of less than 0.05 were considered statistically significant.

Results

General, urinary, and serum variables

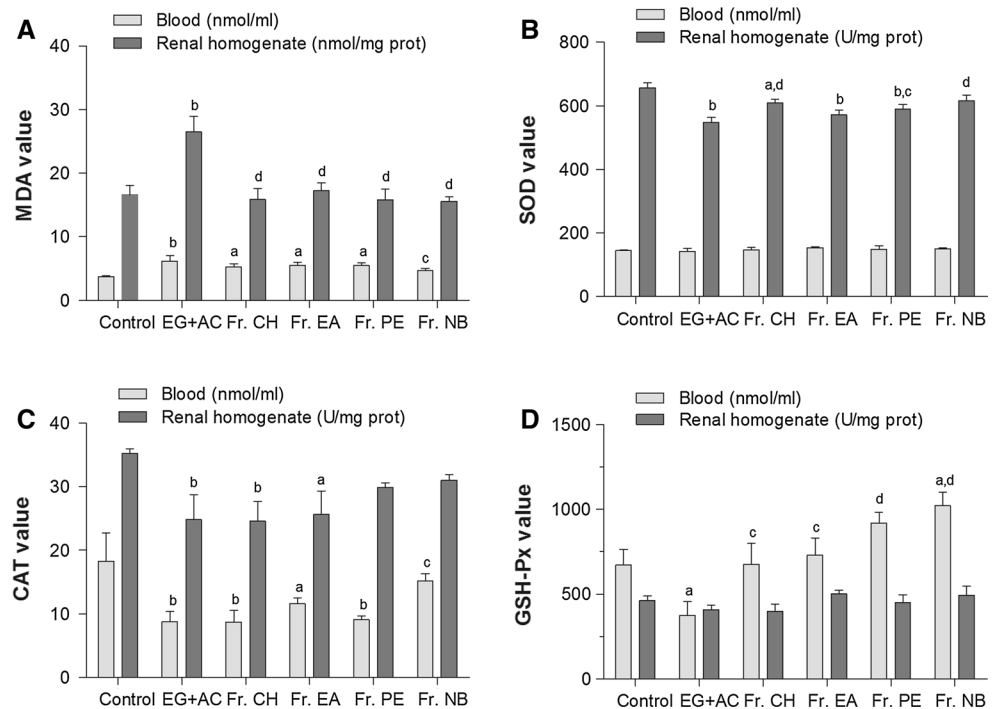
Body weight, water intake, and urinary volume, pH, and composition recorded before the start of treatment were not significantly different among the groups (data not shown). Parameters recorded from groups of animals at the end of 28 days of treatment period are given in Table 2. A significant loss in body weights ($P < 0.01$) by the stone-inducing intervention was observed in lithogenic group during the study period, while in Fr. PE- and Fr. NB-treated groups there were net gains in body weight at the end of the study similar to that of the control animals. However, kidney weights were significantly less in Fr. PE- and Fr. NB-treated groups than the lithogenic group, $P < 0.05$ and $P < 0.01$, respectively. Volume of 24 h urine was higher in lithogenic group compared to those of the normal control animals ($P < 0.05$). Urine pH was also reduced in lithogenic group than normal though not to a significant extent, however, it significantly increased in Fr. PE- and Fr.

Table 2 Parameters recorded from urine and blood of groups after 4 weeks of study

Parameters	Control	EG + AC	Fr. CH	Fr. EA	Fr. PE	Fr. NB
Gain in BW (%)	34.13 ± 1.59	4.29 ± 11.8^a	9.04 ± 4.45^a	12.48 ± 5.02^a	30.97 ± 7.28^b	33.25 ± 8.76^b
Kidney wts (g)	1.17 ± 0.13	1.62 ± 0.33^a	1.38 ± 0.28	1.45 ± 0.41^c	1.24 ± 0.18^d	1.10 ± 0.11^b
Urinary/24 h						
Volume (mL)	31.50 ± 21.17	49.50 ± 14.64^c	45.57 ± 7.25^c	44.25 ± 8.92^c	39.63 ± 4.00	36.86 ± 9.44
pH	7.17 ± 0.51	6.84 ± 0.68	7.45 ± 0.39	7.50 ± 1.19	8.23 ± 0.39^{ab}	7.71 ± 0.72^d
Mg^{2+}	9.39 ± 3.74	4.66 ± 1.21^c	6.03 ± 2.81	6.77 ± 2.81	7.67 ± 4.49	10.17 ± 4.92^b
Ca^{2+}	3.85 ± 5.65	0.94 ± 0.31^c	1.42 ± 1.16	1.27 ± 0.91	2.09 ± 1.48	2.86 ± 1.65
Oxalate	5.28 ± 2.01	11.23 ± 5.14^a	9.94 ± 3.41^c	9.19 ± 2.67^c	7.44 ± 3.96^d	6.93 ± 2.20^d
Citrate	83.60 ± 45.94	39.58 ± 13.63^a	63.53 ± 30.08	60.38 ± 22.05	69.87 ± 13.22^d	73.10 ± 20.39^d
Serum						
Cr	29.13 ± 4.85	59.75 ± 42.59^a	49.17 ± 13.33	47.83 ± 5.74	36.50 ± 12.19^d	30.14 ± 3.63^b
BUN	6.97 ± 1.45	15.55 ± 5.88^a	11.04 ± 1.70^{cd}	11.8 ± 4.64^{cd}	9.4 ± 2.13^b	7.29 ± 1.13^b

^a $P < 0.01$ vs. Control; ^b $P < 0.01$ vs. EG + AC; ^c $P < 0.05$ vs. Control; ^d $P < 0.05$ vs. EG + AC

Fig. 1 Activities of MDA, SOD, CAT, and GSH-Px in blood and kidney homogenates. The values shown are mean $\pm$ SD (8–10). **a** $P < 0.05$ vs. control, **b** $P < 0.01$ vs. control, **c** $P < 0.05$ vs. EG + AC, **d** $P < 0.01$ s. EG + AC



NB-treated groups compared with lithogenic group. Oxalate excretion was also significantly enhanced ($P < 0.01$) in lithogenic group animals whereas Mg^{2+} , Ca^{2+} , and citrate excretion significantly decreased. The serum creatinine and BUN were significantly increased in lithogenic group ($P < 0.01$) as compared to control, however, Fr. PE and Fr. NB treatment significantly lowered the elevated serum levels of creatinine and BUN.

Antioxidant effect of *D. styracifolium* extracts

Figure 1 shows the activities of MDA, SOD, CAT, and GSH-Px in blood and kidney homogenates. Compared to the control group, the MDA values increased significantly, but the CAT values decreased significantly in the lithogenic group both in the blood and kidney samples ($P < 0.01$). Besides, the renal SOD and the blood GSH-Px levels significantly reduced in the lithogenic group ($P < 0.01$ and $P < 0.05$, respectively). The blood SOD and the renal GSH-Px levels slightly reduced in the lithogenic group though not to a significant extent. *D. styracifolium* extracts treatment protected against the changes associated with oxidative stress. Among the four extract fractions, Fr. PE and Fr. NB showed the better antioxidant activities.

Antilithic effect of *D. styracifolium* extracts

The kidneys excised from lithogenic group were enlarged in stone-forming animals, whereas Fr. PE- and Fr. NB-treated groups kidneys were not significantly different from

those of the normal control. When observed under polarized light microscope, in the kidney sections stained by H&E were seen amounts of birefringent crystalline deposits from the histological slices in tubules of all regions, including cortex, medulla, and papilla, of all the animals in the lithogenic group. However, only in 5 out of 10 rats and 4 out of 10 rats treated with Fr. PE and Fr. NB, respectively, were found the crystalline deposits which were also less abundant and visibly smaller (Fig. 2). The renal tubules were also markedly distended with deciduous epithelial cells in the entire kidneys of the rats in lithogenic group (Fig. 3). The crystal deposition scores were 7.5 ± 2.5 , 7.1 ± 2.1 , 6.9 ± 2.6 , 4.0 ± 1.4 , and 2.3 ± 1.4 of lithogenic group, Fr. CH-treated group, Fr. EA-treated group, Fr. PE-treated group, and Fr. NB-treated group, respectively (Fig. 4).

Discussion

Kidney stone formation is a complex process and the result of a cascade of events, including crystal nucleation, growth, aggregation, and retention within the urinary tract [15]. In view of the traditional medical uses of *D. styracifolium* in urolithiasis and the previous studies of *D. styracifolium* crude extract showing the antilithic activity in rats [8–12], we were motivated to further study the extracts from different polarity fractions, including Fr. CH, Fr. EA, Fr. PE, and Fr. NB fractions, of *D. styracifolium* for its antilithic and antioxidant effects in experimental rat model of urolithiasis

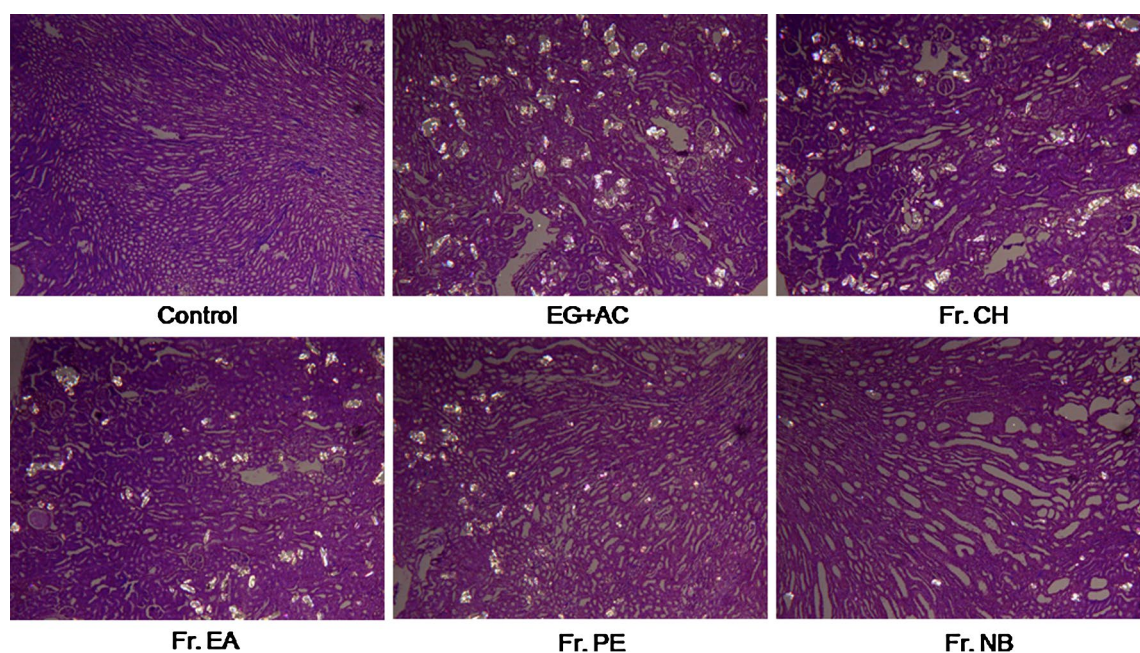


Fig. 2 Representative microscopic images showing the CaOx depositions under polarized light microscopy of the H&E-stained kidney section. The white segments indicate the crystals deposit (magnification $\times 100$)

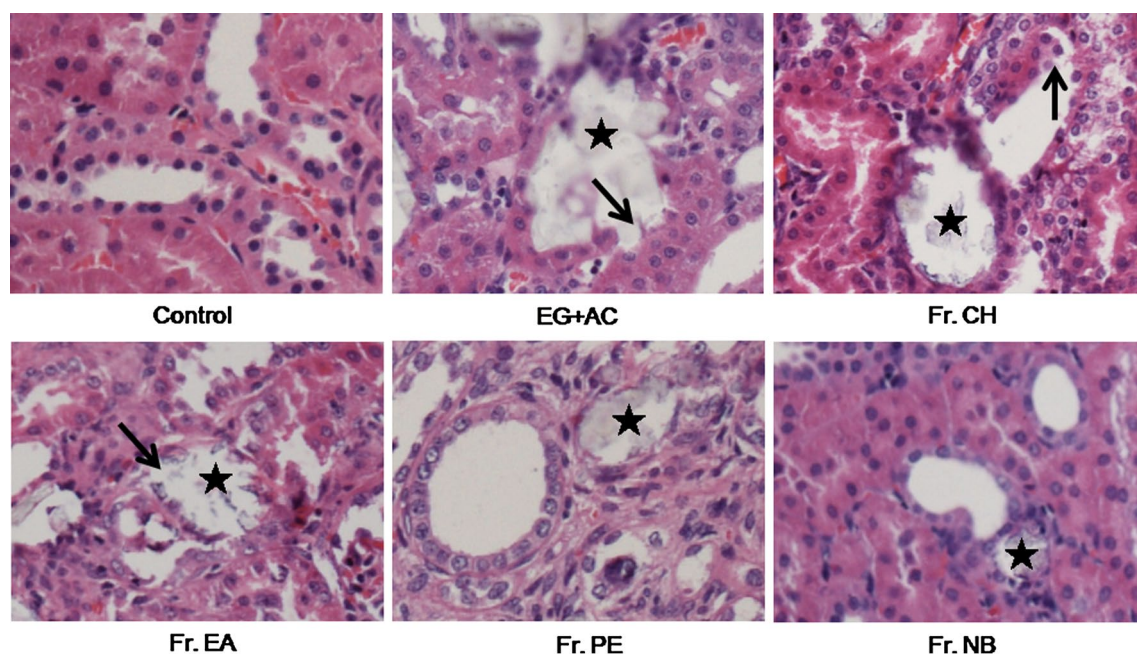


Fig. 3 Representative microscopic images showing the pathological alterations of the H&E-stained kidney section. The asterisks show the crystal deposit, and the arrows show the renal tubular epithelial cells injury (magnification $\times 400$)

to explore the potential phytochemically active constituents of this herb.

Hyperoxaluria is considered to be the major risk factor for CaOx kidney stones formation [16]. Administration of EG to rats is the most frequently used and effective

method to induce renal CaOx deposition in vivo [17]. Other methods include the addition of hydroxy-L-proline to food [18, 19], or administration of AC along with EG to facilitate the formation of CaOx crystals [20, 21]. The primary purpose of these methods is to supply the precursor

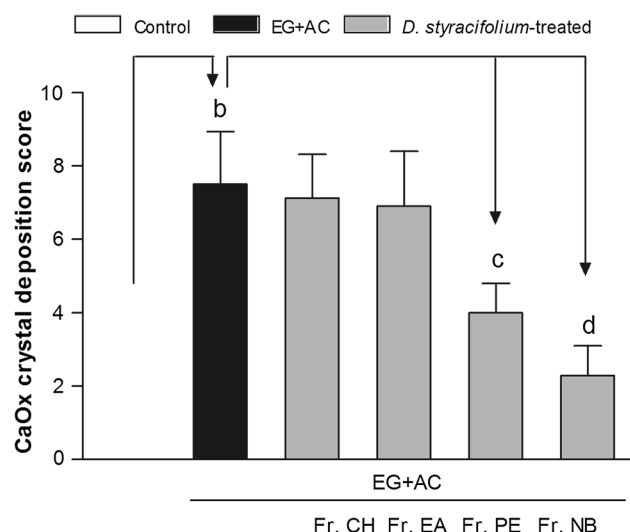


Fig. 4 CaOx crystal deposition score at the end of the study. The data shown are mean $\pm$ SD ($n = 10$). *b* $P < 0.01$ vs. control, *c* $P < 0.05$ vs. EG + AC, *d* $P < 0.01$ vs. EG + AC

of oxalate as well as to cause oxidative stress and damages in the renal epithelial cells. In the present study, 1 % EG + 2 % AC orally induced hyperoxaluria rat model of experimental kidney stone was used [13]. Hyperoxaluria was successfully induced since urinary excretion of oxalate increased significantly, and the amount of crystals deposit in the kidney increased as well concomitantly. It is reported that herbal extracts may influence the oxalate metabolism by decreasing the formation of oxalate synthesizing enzymes and lactate dehydrogenase, thus decreasing the excretion of urinary oxalate and renal crystal deposits [22]. Therefore, decrease in excretion of oxalate may explain the prevention from oxidative stress and renal crystal deposition.

It is considered, from the theory of traditional Chinese medicine, that moistness and heat are the pathogenesis of urolithiasis. Due to its heat-clearing and diuretic properties, *D. styracifolium* (Osbeck) Merr. has been traditionally used for treatment of renal stones since antiquity [23]. On the basis of these facts, we report herein the antilithic and antioxidant effects of various extracts of *D. styracifolium*. Our data revealed that kidney CaOx crystals deposits in Fr. PE- and Fr. NB-treated groups were significantly lower than those in the lithogenic group, Fr. CH- and Fr. EA-treated groups, suggesting that the antilithic effect of Fr. PE and Fr. NB are stronger than Fr. CH and Fr. EA. Although the structure and bioactive components responsible for the antilithic effects of these two bioactive fractions are not fully elucidated, phytochemical constituents such as flavonoid, alkaloids, phenols, tannin, polysaccharide, etc., are present in *D. styracifolium*, which are believed to be responsible for the biological and pharmacological

effects. Consistent with the antilithic activity in the kidney, the urinary excretion of oxalate was reduced concomitantly. These data provide a further rationale for the medical use of *D. styracifolium* in urolithiasis from the level of constituents of different polarity fractions, and meanwhile invoke an interesting question that what are the active compounds extracted from Fr. PE and/or Fr. NB fractions of *D. styracifolium*.

Studies of animal models as well as tissue culture have indicated that injury to the renal tubular epithelial cells in the presence of oxalate and/or CaOx crystals leads to the production of reactive oxygen species, which mainly consist of the superoxide anion ($O_2^{\bullet-}$), nitric oxide radical ($NO^{\bullet}$), hydroxyl radical ($OH^{\bullet}$), and hydrogen peroxide (H_2O_2) [24]. $O_2^{\bullet-}$ is highly reactive, therefore, it is acted on by the scavenging enzyme SOD, which converts it to H_2O_2 . The H_2O_2 is subsequently metabolized to water via CAT or by GSH-Px in the presence of reduced glutathione [25]. In vivo and in vitro studies have demonstrated that hyperoxaluria results in production of superoxide and hydroxyl free radicals causing oxidative stress, cell membrane rupture, and cell injury, and leads to CaOx crystal adherence and retention in renal tubules [26–28]. Several studies have shown that antioxidants can protect against oxidative injury by oxalate and crystal deposition [29, 30]. From this it can be speculated that the inhibitory effect of *D. styracifolium* on CaOx crystal deposition in renal tubules could have also been due to its antioxidant activity. In this study, we demonstrated the oxidative damage of untreated lithogenic rats accompanied by extensive calcium CaOx deposition in kidneys, as reflected from increased level of MDA, the marker of peroxidative injury, and decreased activities of antioxidant enzymes as well as deteriorated renal functions. The renal tubules were markedly distended with deciduous epithelial cells in the entire kidney. The inhibitory effect of Fr. PE and Fr. NB on CaOx retention in the kidneys thus could have also been due to its antioxidant activity.

In conclusion, our results show that the *D. styracifolium* extracts contain several active compounds which may be beneficial for inhibiting the formation of CaOx. The Fr. PE and Fr. NB of *D. styracifolium* effectively prevent kidney stone formation probably through inhibition of calcium oxalate crystallization and antioxidant activity. Research will be continued to elucidate the mechanism of the activities and particularly the structure–activity in the most bioactive fractions.

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Conflict of interest The authors declare no conflict of interest.

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